17 Here are the first five terms of an arithmetic sequence.

2 5 8 11 14

Prove algebraically that the sum of the squares of any two consecutive terms of this sequence is always 1 less than a multiple of 6

Consecutive terms are $3n - 1$ and $3n + 2$.

1 mark
Find general terms.

Sum of squares = $(3n - 1)^2 + (3n + 2)^2$

1 mark
Square both terms.

$= 9n^2 - 6n + 1 + 9n^2 + 12n + 4$

$= 18n^2 + 6n + 5$

1 mark
Expand and simplify.

$= 6(3n^2 + n + 1) - 1.$

For integer n, $3n^2 + n + 1$ is an integer.

1 mark
Factor 6; explain why the bracket is integer.

So the sum is one less than a multiple of 6.

(Total for Question 17 is 4 marks)

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